

which makes them a good match for the application.

Bendable, twistable electronic circuits whose performance nearly matches that of conventional CMOS chips, have been reported. These circuits, developed by a team of researchers at the University of Illinois at Urbana-Champaign (USA), are built from ribbons of silicon only a few nanometres thick that are mounted on flexible plastic substrates. The same group of researchers has developed an improved plastic circuit that is not only flexible but also stretchable and foldable. To make it foldable, they looked at the behaviour of everyday objects and decided to make the circuit much thinner.

The researchers made plastic circuits by transferring thin ribbons of silicon onto glue-coated plastic using a patterned rubber stamp. But before the ultra-thin silicon layer is applied to the substrate, the plastic is heated, causing it to expand. Once the circuit layer is deposited and chemically bonded to the expanded substrate, the plastic is

allowed to cool and contract. Relaxing the strain causes the circuit layer to buckle and form wavy patterns like the bellows of an accordion. It's the folds and wrinkles that give the circuit the ability to stretch and bend without breaking. The researchers said that in laboratory tests, the circuits showed no signs of fatigue even after a few hundred stretch-and-release cycles.

Applications abound

Flex circuits are often used as connectors in various applications where flexibility, space savings or production constraints limit the serviceability of rigid circuit boards or hand wiring. In addition to cameras, a common application of flex circuits is in computer keyboard manufacturing; today, most keyboards use flex circuits for the switch matrix.

Flexible circuits are advantageous in numerous applications:

1. Tightly assembled electronic packages, where electrical connections are required in three axes, such as cameras (static application)

2. Electrical connections where the assembly is required to flex during its normal use, such as folding cell phones (dynamic application)

3. Electrical connections between sub-assemblies to replace wire harnesses, which are heavier and bulkier, such as in rockets and satellites

4. Electrical connections where board thickness or space constraints are driving factors

The researchers are entering into partnerships with physicians for development of flexible biomedical devices that can be implanted in the human body in a more comfortable manner; for example, an implantable sensor that will monitor electrical activity in the brain to help predict the onset of epileptic seizures. The device may also work in reverse, sending electric pulses that head off the seizures. This requires a device that will conform to the rippled geometry of the brain. ●

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